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Neptun-code: HGRU8P

Part 2 of the design project

Determine the factor of safety against sliding of the embankment with the slices method (simplified Bishop), and for the cut with the block method. With the method of slices, consider 3 different slip circles (base failure, slope failure with positive tangent and slope failure with negative tangent at the slope toe).

According to the design approach DA-3 of Eurocode 7, use the design values of the soil parameters to determine the safety factor.

Top width of the road:	12 m
Inclination of the pavement:	3,0 %
Height of the embankment (in axis):	9,2 m
Slope of the embankment:	5/4
Height of the cut (in axis):	12,7 m
Slope of cut:	6/4

Characteristic values of soil parameters for the embankment:

Unit weight:	18,0 kN/m ³
Internal friction angle:	28°
Cohesion:	15 kPa

Characteristic values of soil parameters in cut:

Layer 1 unit weight:	20 kN/m ³
Layer 1 internal friction angle:	26°
Layer 1 cohesion:	5 kPa
Layer 1 height:	H/4

Layer 2 unit weight:	19,0 kN/m ³
Layer 2 internal friction angle:	30°
Layer 2 cohesion:	15 kPa
Layer 2 height:	H/2
Inclination of the slip plane:	9°

Characteristic values of shear strength on the slip surface:

Internal friction angle in the slip surface:	10°
Cohesion in the slip surface:	10 kPa

Teacher notifications:

	<i>Date</i>	<i>Signature</i>
<i>Consultation</i>		
<i>Submission</i>		